

Is Costa Rica's Transition Fast Enough for 1.5°C?

CO₂, GDP and Transport in a Supposed Green Champion

Group PG-20

Plan

1. Q1 – Are we on a 1.5°C path?

If we assume a simple linear decline in CO₂ per capita from 2010 to net-zero by 2050, how far is Costa Rica above or below that pathway today?

2. Q2 – Decoupling compared with similar economies

Among countries that had similar GDP per capita to Costa Rica in 2010, has Costa Rica done unusually well (or not) at growing income while stabilising or reducing CO₂ per capita?

3. Q3 – Is transport now the main bottleneck?

Given that Costa Rica's electricity is already mostly low-carbon, to what extent are transport emissions now dominating its remaining CO₂ and limiting further progress?

4. We then discuss what this implies for whether Costa Rica's transition is fast enough and where the main bottlenecks lie.

Data & approach

- **Country focus:** Costa Rica
- **Core data:** CO₂ emissions, CO₂ per capita, GDP per capita, and sectoral emissions
- **Source:** Our World in Data – CO₂ and Energy
- **Period:** From 1960s to early 2020s, with a focus on 2010 onwards for the 1.5°C pathway

Q1 – 1.5°C pathway

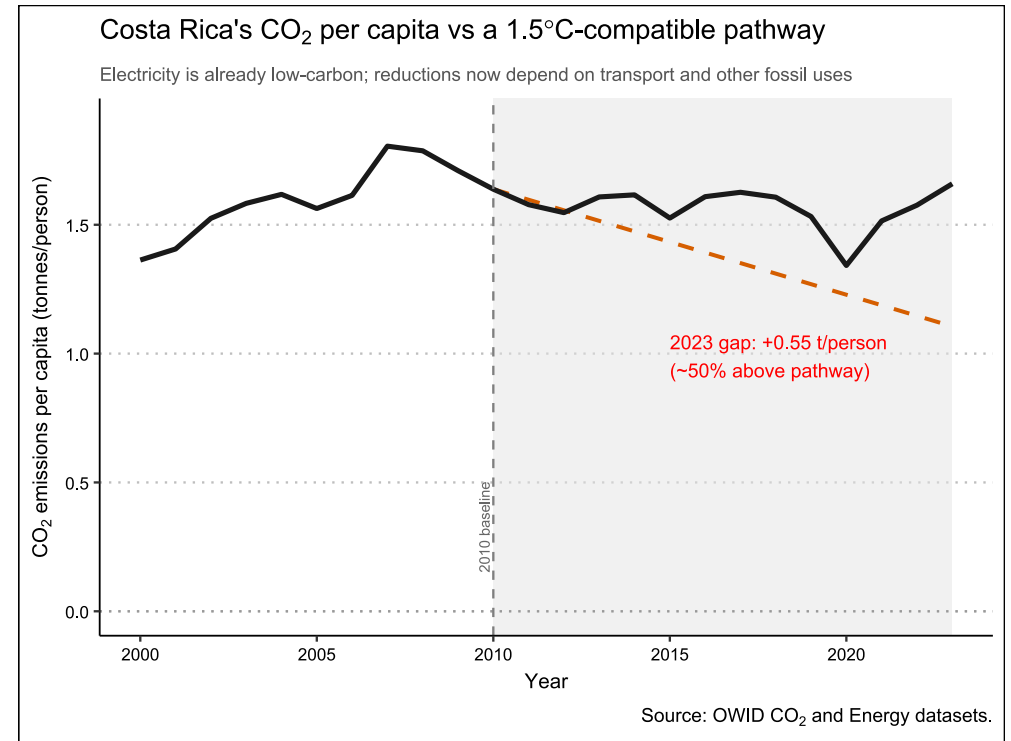
Compare observed CO₂ per capita with a **simple linear net-zero-by-2050 path** starting in 2010.

Q2 & Q3 – Structure and sectors

- Build a **peer group** of countries with similar GDP per capita in 2010
- Compare **GDP vs CO₂ per capita changes** and then zoom in on **transport vs other sectors** for Costa Rica

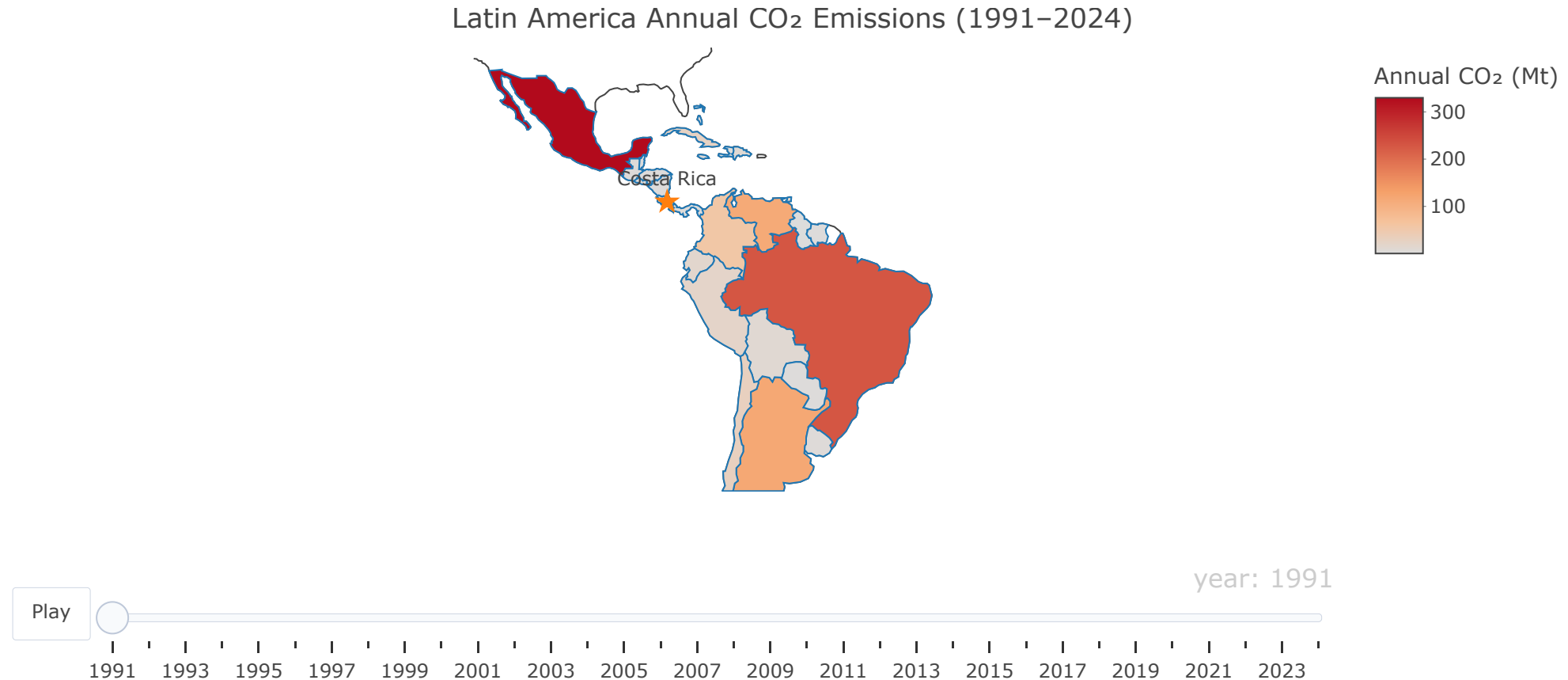
Q1 – Are we on a 1.5°C path?

- I use a **simple stylised pathway**: CO₂ per capita declines **linearly** from 2010 to **net-zero** in 2050.
- Costa Rica's **observed CO₂ per capita** is then plotted against this pathway.
- The **gap in the latest year** shows how far Costa Rica is **above** the 1.5°C-style path, under these assumptions.
- This is not a full carbon budget model – it's a **transparent, back-of-the-envelope benchmark**.



Latin America CO₂ context

Brazil and Mexico dominate regional totals; Costa Rica sits in the low-to-mid range of emitters.



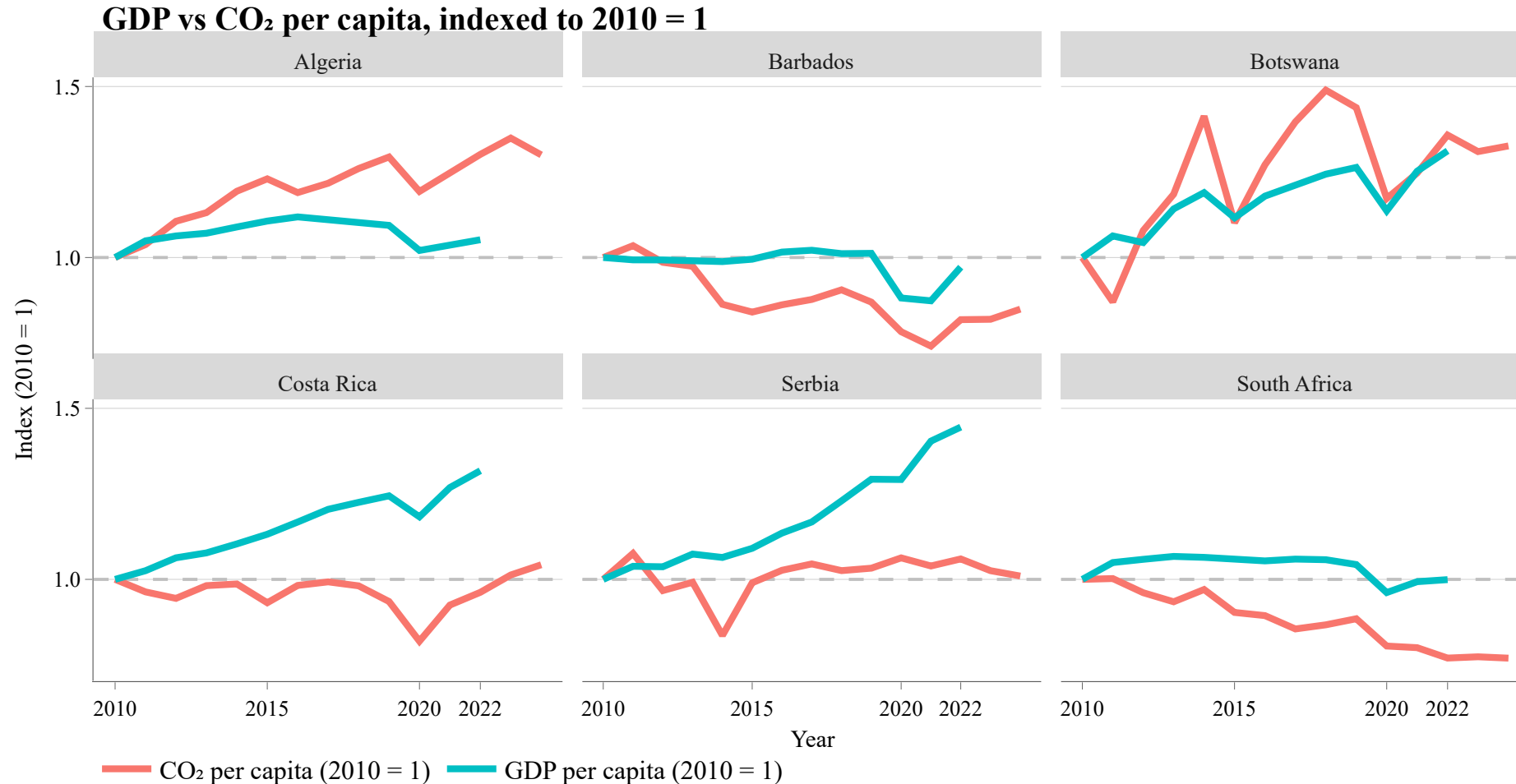
Q2 – GDP vs CO₂ change since 2010

- Here I summarise **total change from 2010 to the latest year** for each country.
- **GDP_pc_change**: growth in income per person.
- **CO₂_pc_change**: change in CO₂ per person.
- Costa Rica should stand out as **strong GDP growth with flat or slightly falling CO₂**, whereas some peers grow both.
- Costa Rica combines **~32% GDP per capita growth** with slightly falling CO₂, whereas some peers trade higher emissions for similar or lower growth.

Change in GDP and CO₂ per capita since 2010

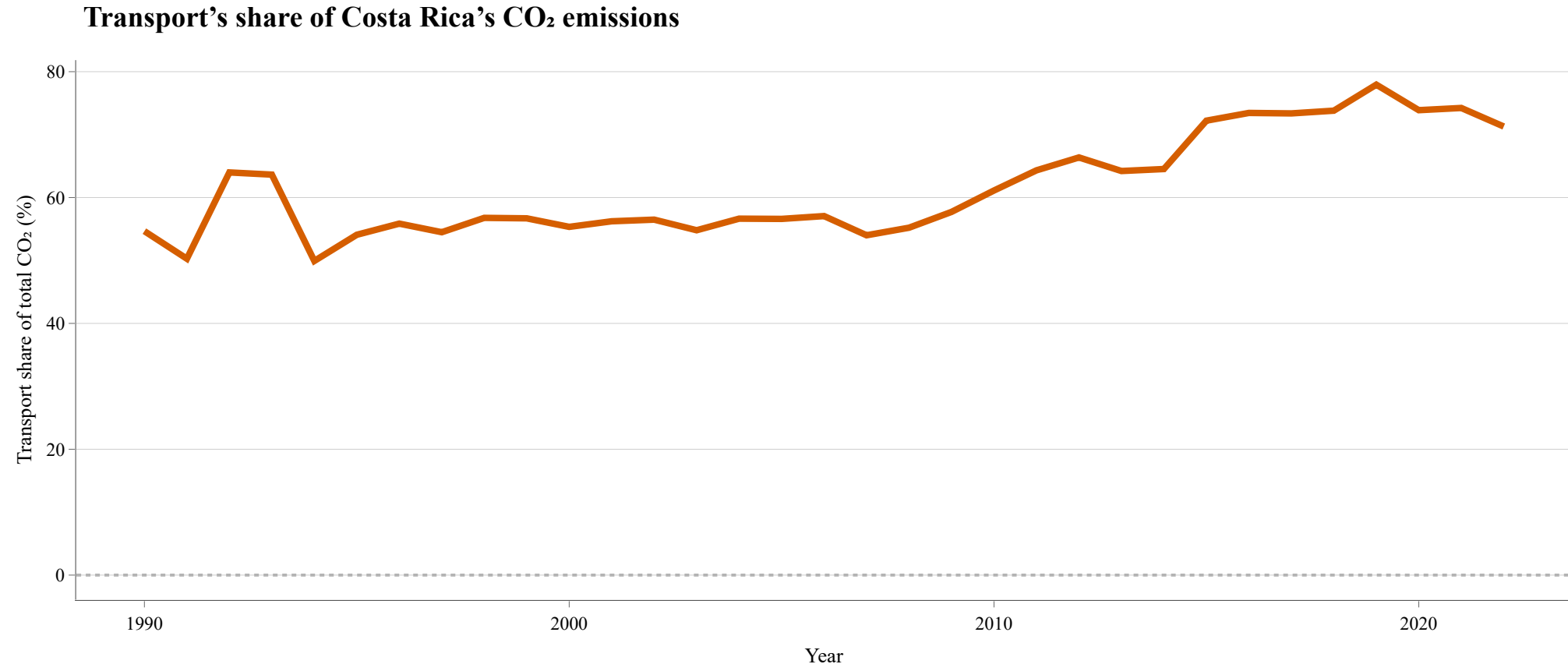
country	Start year	End year	GDP % change	CO2 % change
Serbia	2010	2022	+44.5%	+6.0%
Costa Rica	2010	2022	+31.8%	-3.8%
Botswana	2010	2022	+31.2%	+35.8%
Algeria	2010	2022	+5.2%	+30.2%
South Africa	2010	2022	-0.1%	-23.0%
Barbados	2010	2022	-2.8%	-18.2%

Q2 – Decoupling compared with similar economies



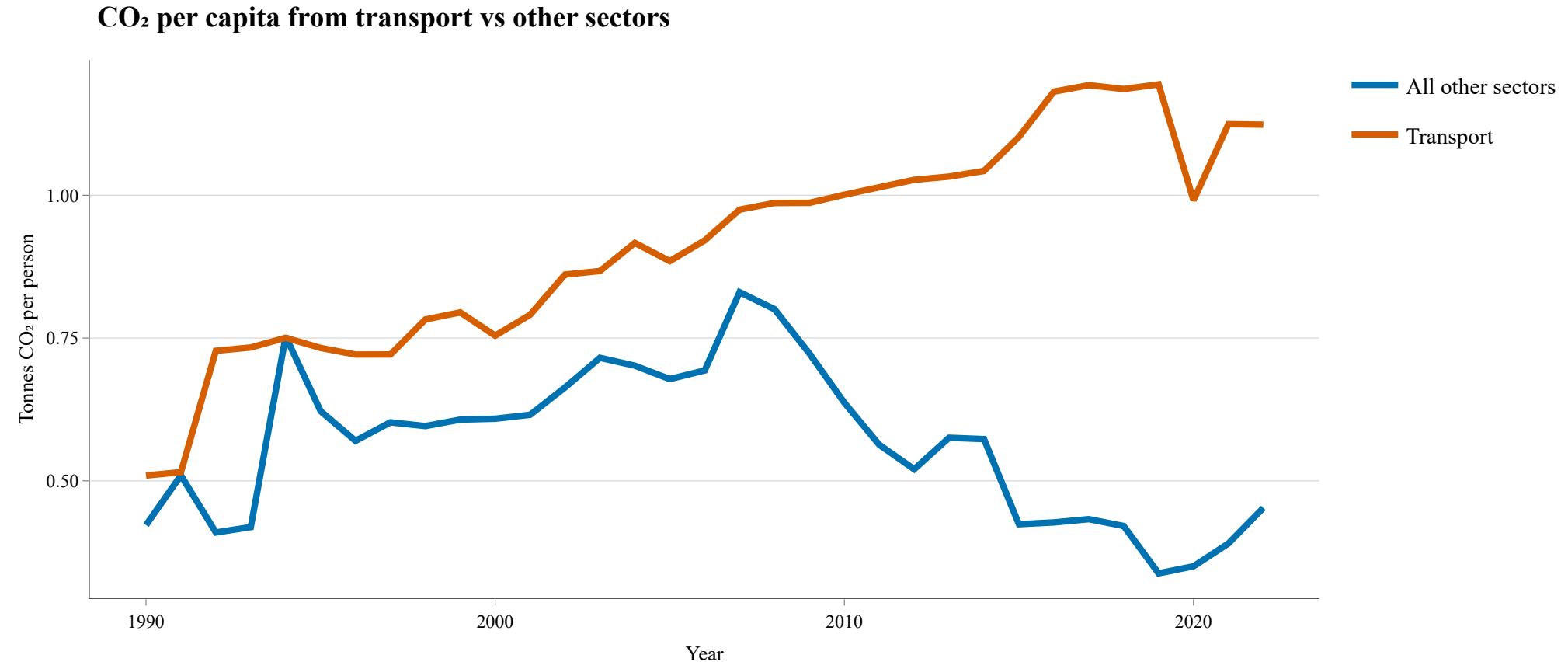
Q3 – Has transport become the main bottleneck?

Transport rises to around 71-75% of national CO₂, meaning almost all remaining emissions are now transport-related.



Q3 – CO₂ per capita: transport vs other sectors

Other sectors decline or flatten; transport hardly falls, making it the main brake on further progress



Discussion

- **Q1 – 1.5°C pathway**

Under a simple “net-zero by 2050” linear pathway starting in 2010, Costa Rica’s CO₂ per capita in 2023 sits around **50% above** where it would need to be. That means future cuts must be **steeper than the past trend** if it is to realign with a 1.5°C-type path.

- **Q2 – Relative decoupling, but not a miracle**

Compared with countries that had similar GDP per capita in 2010, Costa Rica manages **strong income growth with flat or slightly falling CO₂ per capita**, while some peers grow both GDP and emissions and others cut emissions at the cost of weaker growth. It looks **better than a typical peer**, but it is **not alone** in reducing emissions.

- **Q3 – Transport is now the binding constraint**

Electricity is already largely low-carbon, so total CO₂ no longer falls unless **transport emissions start to drop**. Transport’s share of emissions has climbed to the **majority of Costa Rica’s CO₂**, and transport CO₂ per capita has not fallen nearly as fast as other sectors. That makes **transport the main bottleneck** to a faster transition.

- **Limitations and robustness**

Our pathway is deliberately simple (linear, national-average CO₂ per capita, no full carbon budget). Results are sensitive to the **choice of baseline year, peer group, and whether we use production-based or consumption-based emissions**, but under reasonable choices the core picture (above pathway; moderate decoupling; transport bottleneck) is stable.

Conclusion – Is Costa Rica's transition fast enough for 1.5°C?

- **Short answer:** Under our simple 1.5°C-style pathway, **no** – Costa Rica is currently **overshooting** its implied per-capita emissions level by roughly half, so “carrying on as usual” is **not fast enough** to close the gap.
- **Relative performance:** Compared with similar-income economies since 2010, Costa Rica looks **relatively well-behaved** – it has managed to grow richer without increasing CO₂ per person – but it is **not yet on a path** consistent with rapid global decarbonisation.
- **What needs to change:** With electricity mostly decarbonised, **further progress depends on transport:** accelerating electrification of vehicles, cleaner fuels and modal shift. Without a step-change in transport policy and infrastructure, Costa Rica will remain **a green champion in branding** but not in terms of a **1.5°C-compatible emissions trajectory**.

